

Project report

Project Title

RFID Access Control System with excel data storing

Objective

The primary objective of this project is to develop an RFID-based access control system. The system uses RFID technology to identify and verify users. Upon successful identification, the system grants access by unlocking a door via a servo motor. The system also logs the access events, including the date and time of entry, and the UID and name of the user.

Components Used

- **MFRC522 RFID Reader:** Used to read the UID of RFID cards.
- **DS3231 RTC Module:** Real-time clock module to keep track of date and time.
- **Servo Motor:** Controls the locking mechanism of the door.
- **LEDs and Buzzer:** Provide visual and auditory feedback.
- **Arduino Board:** Controls the entire system.

Hardware Configuration

Components and Connections:

MFRC522 RFID Reader

- **VCC:** 3.3V on the Arduino
- **GND:** GND on the Arduino
- **SCK:** Pin 13 (SCK on the Arduino)
- **MOSI:** Pin 11 (MOSI on the Arduino)
- **MISO:** Pin 12 (MISO on the Arduino)
- **SS:** Pin 10
- **RST:** Pin 9

RTC Module (DS3231)

- **VCC:** 5V on the Arduino
- **GND:** GND on the Arduino

- **SDA:** Pin A4
- **SCL:** Pin A5

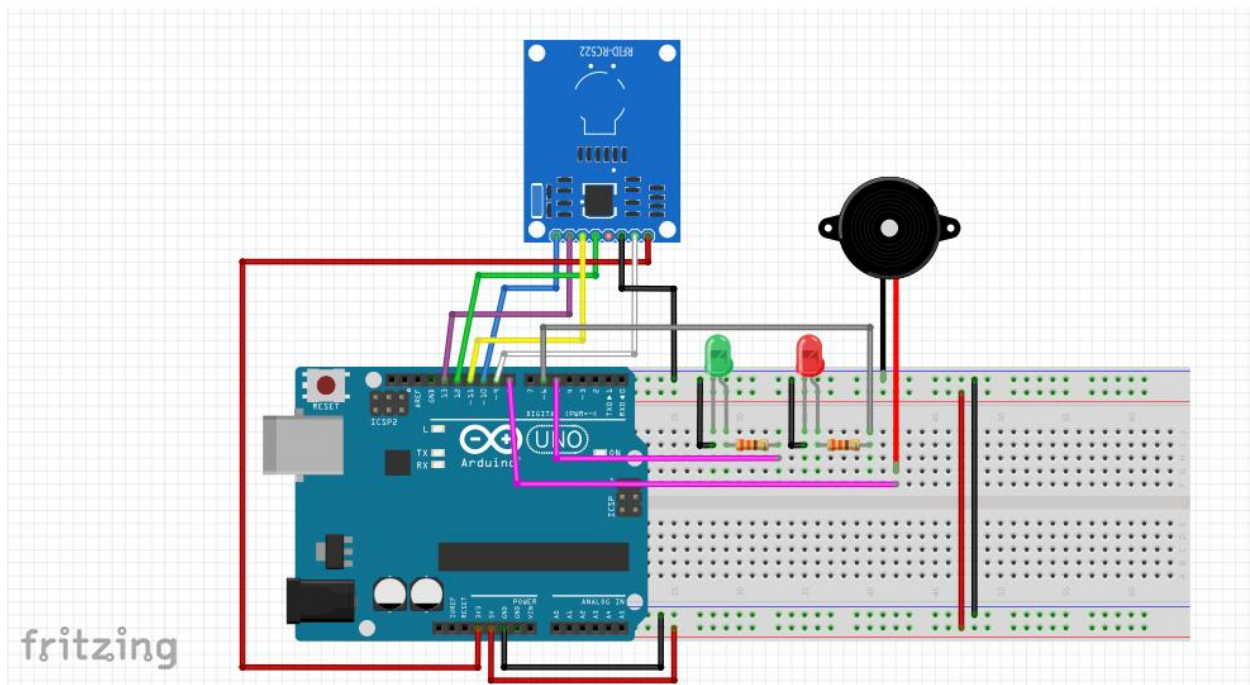
Servo Motor

- **VCC:** External 5V power supply (Make sure to connect the GND of this power supply to the Arduino GND)
- **GND:** Common GND with the Arduino
- **Signal:** Pin 3

LEDs and Buzzer

- **Red LED**
 - Anode: Pin 6
 - Cathode: GND (via a 220-ohm resistor)
- **Green LED**
 - Anode: Pin 5
 - Cathode: GND (via a 220-ohm resistor)
- **Buzzer**
 - Positive: Pin 8
 - Negative: GND

CIRCUIT DIAGRAM:



Libraries Used

- **SPI.h:** Used for communication with the RFID reader.
- **MFRC522.h:** Specific library for interfacing with the MFRC522 RFID reader.
- **Wire.h:** Used for I2C communication with the RTC module.
- **RTCLib.h:** Provides functions to interact with the DS3231 RTC module.
- **Servo.h:** Provides functions to control the servo motor.

Functions

1. **setup()**
 - Initializes the serial communication.
 - Initializes the SPI bus and the RFID reader.
 - Initializes the RTC module.
 - Sets the initial position of the servo motor (locked).
 - Configures the pins for the LEDs and the buzzer.
 - Prints the header for the data logging.
2. **loop()**
 - Continuously looks for a new RFID card.
 - Reads the UID of the card if present.
 - Checks if the UID matches predefined UIDs.
 - Logs the access event with the current date, time, UID, name, and number.
 - Provides visual and auditory feedback.
 - Unlocks the door if the UID is recognized.
 - Keeps the door unlocked for 5 seconds, then locks it again.
3. **printUID()**
 - Prints the UID of the detected RFID card.
4. **beepBuzzer()**
 - Beeps the buzzer thrice as an indication of an unrecognized UID.

Code Listing

Cpp

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <RTCLib.h>
#include <Servo.h>

#define SS_PIN 10 //RX slave select
#define RST_PIN 9

MFRC522 mfr522(SS_PIN, RST_PIN); // Create MFRC522 instance.
RTC_DS3231 rtc;
Servo myServo;

byte card_ID[4]; //card UID size 4byte
byte Name1[4] = {0x63, 0xF0, 0x0A, 0x0E}; //first UID card
```

```

byte Name2[4] = {0xC3, 0x8F, 0x06, 0x10}; //second UID card

int const RedLed = 6;
int const GreenLed = 5;
int const Buzzer = 8;
int const ServoPin = 3;

String Name; //user name
long Number; //user number

void setup() {
  Serial.begin(9600); // Initialize serial communications with the PC
  SPI.begin(); // Init SPI bus
  mfrc522.PCD_Init(); // Init MFRC522 card

  if (!rtc.begin()) {
    Serial.println("Couldn't find RTC");
    while (1);
  }

  if (rtc.lostPower()) {
    Serial.println("RTC lost power, setting the time!");
    rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  }

  myServo.attach(ServoPin);
  myServo.write(0); // Initial position (locked)

  Serial.println("CLEARSHEET"); // clears starting at row 1
  Serial.println("LABEL,Date,Time,UID,Name,Number");// make five columns
  (Date,Time,UID,[Name:"user name"],[Number:"user number"])

  pinMode(RedLed, OUTPUT);
  pinMode(GreenLed, OUTPUT);
  pinMode(Buzzer, OUTPUT);
}

void loop() {
  //look for new card
  if (!mfrc522.PICC_IsNewCardPresent()) {
    return; //go to start of loop if there is no card present
  }
  // Select one of the cards
  if (!mfrc522.PICC_ReadCardSerial()) {

```

```
    return; //if read card serial(0) returns 1, the uid struct contains the ID of
the read card.
}
```

```
printUID(); // Print the UID of the card
```

```
for (byte i = 0; i < mfrc522.uid.size; i++) {
    card_ID[i] = mfrc522.uid.uidByte[i];
```

```
    if (card_ID[i] == Name1[i]) {
        Name = "First Employee"; //user name
        Number = 123456; //user number
    } else if (card_ID[i] == Name2[i]) {
        Name = "Second Employee"; //user name
        Number = 789101; //user number
    } else {
        digitalWrite(GreenLed, LOW);
        digitalWrite(RedLed, HIGH);
        beepBuzzer(); // Call the function to beep the buzzer
        goto cont; //go directly to the labeled section
    }
}
```

```
DateTime now = rtc.now();
Serial.print("DATA,");
Serial.print(now.year(), DEC); Serial.print('/');
Serial.print(now.month(), DEC); Serial.print('/');
Serial.print(now.day(), DEC); Serial.print(",");
Serial.print(now.hour(), DEC); Serial.print(':');
Serial.print(now.minute(), DEC); Serial.print(':');
Serial.print(now.second(), DEC); Serial.print(",");
Serial.print(" ");
for (byte i = 0; i < 4; i++) {
    Serial.print(card_ID[i], HEX);
    if (i < 3) Serial.print(":");
}
Serial.print(",");
Serial.print(Name); //send the Name to excel
Serial.print(",");
Serial.println(Number); //send the Number to excel
digitalWrite(GreenLed, HIGH);
digitalWrite(RedLed, LOW);
digitalWrite(Buzzer, HIGH);
delay(30);
digitalWrite(Buzzer, LOW);
```

```

myServo.write(90); // Unlock position
delay(5000); // Keep the door unlocked for 5 seconds
myServo.write(0); // Lock position
Serial.println("SAVEWORKBOOKAS,Names/WorkNames");

delay(1000);
cont:
delay(2000);
digitalWrite(GreenLed, LOW);
digitalWrite(RedLed, LOW);
}

void beepBuzzer() {
  for (int i = 0; i < 3; i++) {
    digitalWrite(Buzzer, HIGH);
    delay(100);
    digitalWrite(Buzzer, LOW);
    delay(100);
  }
}

void printUID() {
  Serial.print("UID:");
  for (byte i = 0; i < mfrc522.uid.size; i++) {
    Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
    Serial.print(mfrc522.uid.uidByte[i], HEX);
  }
  Serial.println();
}

```

Conclusion

The RFID Access Control System successfully reads RFID cards, checks for authorized users, logs access events, and controls access using a servo motor. The system provides feedback using LEDs and a buzzer, making it user-friendly and efficient. SAVES DATA IN EXCEL SHEET

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